CRIME REPORT MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

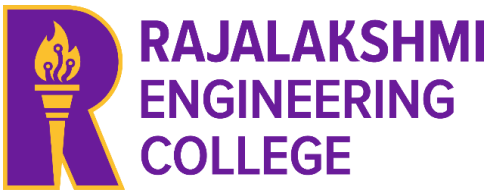
ASHWANTH K (240701057)

in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

BONAFIDE CERTIFICATE

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Certified that this project report titled "Crime Report Management System" is the bonafide work of **ASHWANTH K** (240701057), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE	**SIGNATURE**
Dr.J.Manoranjini **HEAD OF THE DEPARTMENT** Department of Computer Science And Engineering Rajalakshmi Engineering College Thandalam – 602 105	Dr. Jinu Sophia **COURSE FACULTY** Assistant Professor Department of Computer Science And Engineering Rajalakshmi Engineering College Thandalam – 602 105

Submitted for the End Semester Examination (Individual Project) held on ____ Nov, 2026

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, ASHWANTH K (240701057), hereby declare that the project report titled "Crime Report Management System", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, dashboard, user profiles, roles, sharing rules and device layouts — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

Date: ____-11-2026

SIGNATURE OF THE STUDENT
ASHWANTH K
240701057

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman THIRU.S.MEGANATHAN, Chairperson Dr.(Mrs.) THANGAM MEGANATHAN and Vice-Chairman Mr. ABHAY SHANKAR MEGANATHAN for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr.S.N.MURUGESAN, Rajalakshmi Engineering College, for the constant encouragement given to students to take up industry-relevant learning.

I thank Dr.J.MANORANJINI, Head of the Department, Department of Computer Science and Engineering, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, Dr.JINU SOPHIA, Assistant Professor, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and its learning resources, and I thank my family and friends for their constant support.

ASHWANTH K

ABSTRACT

Police departments receive crime complaints every day, and each complaint passes through many steps: it is recorded, given to an officer, investigated, supported with evidence, resolved and closed. In many places these steps are still handled with paper registers and separate files. Information is written again and again, a complaint is not easily connected to the officer, the investigation and the evidence, supervisors cannot see how many cases are in each status, and citizens have no simple way to give feedback. This slows down the work and makes supervision difficult.

The Crime Report Management System is a low-code application built on Zoho Creator to solve this problem. It manages the crime report life cycle in eight connected forms: Citizen_Registration, Crime_Report_Form, Police_Officer, Case_Assignment, Investigation_Details, Evidence_Management, Case_Resolution and Citizen_Feedback. The forms are connected through picklist references, so a report, an officer or a citizen is selected from existing records instead of being typed again. Address fields capture the location of the crime with latitude and longitude, and evidence files can be uploaded with the report and with the evidence record. Eight reports, one for every form, let users view and manage the records. The Admin_Dashboard shows crime reports by type and cases by status with two donut charts and a bar chart, and it embeds the case-assignment form and the crime-report register. Seven profiles, the roles CEO, Police Officer and Citizen, and five sharing rules control who can see which records. A section-based Space menu organises the modules, and phone and tablet layouts are available for all eight forms.

The application replaces paper registers with one online system, reduces repeated data entry, makes every case traceable through its Report_ID, and gives administrators a live view of crime categories, case progress and report information. It was developed entirely through configuration, without custom scripts, which shows how quickly a useful business application can be produced on a low-code platform.

Keywords: Low-Code Development; Zoho Creator; Crime Report Management; Case Assignment; Evidence Management; Dashboard; Role-Based Access Control

PROJECT AT A GLANCE

Attribute	Details
Project Title	Crime Report Management System
Application Domain	Public safety / Police administration – crime complaint, case assignment, investigation, evidence, resolution and feedback
Student Name & Register No.	ASHWANTH K – 240701057
Zoho Creator Edition	[Trial / Education – mention the edition used]
Application Name (Link Name)	Crime_report_management_system
Published / Live URL	[paste the live application URL here]
Demo Credentials (per role)	[Administrator | Police Officer | Citizen – add user names only; never write passwords here]
Number of Forms / Subforms	8 / 0
Number of Reports (by type)	List: 8 Chart: 0 (three charts are placed on the dashboard) Kanban: 0 Calendar: 0 Pivot: 0
Number of Pages/Dashboards	1 (Admin_Dashboard: 3 charts, 1 embedded form, 1 embedded report)
Number of Workflows	Form: 0 Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0 (configuration-based application)
Deluge Scripts / Functions	0 / 0
Integrations	None (built-in address coordinate capture and file upload are used)
Roles & Profiles Defined	Roles used: CEO, Police Officer, Citizen. Profiles: Administrator, Police_Officer_Permissions, Citizen_Permissions, Read, Write, Developer, Customer. Sharing rules: 5
Device Layouts	Web (Lato font, icon text CRM); Phone (all 8 forms, two-column, Chrysler Purple); Tablet (all 8 forms, sliding pane, Dark Grey)
Environments Used	Development → Production (strategy described in Chapter 7)
Date of Production Deployment	[dd-mm-2026]

TABLE OF CONTENTS

BONAFIDE CERTIFICATE ii
DECLARATION iii
ACKNOWLEDGEMENT iv
ABSTRACT v
PROJECT AT A GLANCE vi
LIST OF TABLES ix
LIST OF FIGURES x
LIST OF ABBREVIATIONS xi
CHAPTER 1 INTRODUCTION 1
1.1 Background 1
1.2 Problem Statement 1
1.3 Objectives 2
1.4 Scope of the Project 2
1.5 Justification for a Low-Code Approach 2
1.6 Organisation of the Report 3
CHAPTER 2 LITERATURE SURVEY AND SYSTEM STUDY 4
2.1 Study of the Existing System 4
2.2 Limitations of the Existing System 4
2.3 Low-Code versus Traditional Development 5
2.4 Literature Survey 5
2.5 Emerging Trends in Low-Code Development 6
2.6 Proposed System 6
CHAPTER 3 REQUIREMENT ANALYSIS 7
3.1 Stakeholders and User Roles 7
3.2 Functional Requirements 7
3.3 Non-Functional Requirements 8
3.4 Hardware and Software Requirements 8
3.5 Use Case Model 9
CHAPTER 4 SYSTEM DESIGN 11
4.1 System Architecture 11
4.2 Data Model 11
4.3 Form Design 13
4.4 Process and Workflow Design 17
4.5 User Interface and Navigation Design 18
4.6 Access Control Design 18
CHAPTER 5 IMPLEMENTATION 20
5.1 Application Setup 20
5.2 Forms and Data Capture 20
5.3 Reports 21
5.4 Pages and Dashboards 22

5.5 Scripting and Workflow Automation 23
5.6 Integrations and Built-in Platform Features 23
5.7 User Management and Role-Based Access 23
5.8 AI and Analytics Features 24
5.9 Pro-Code Extension 24
5.10 Mobile and Tablet Access 24
CHAPTER 6 TESTING AND DEBUGGING 26
6.1 Testing Strategy 26
6.2 Test Cases 26
6.3 Debugging Approach 27
6.4 Defect Log 28
6.5 User Acceptance Testing 28
CHAPTER 7 DEPLOYMENT, SECURITY AND GOVERNANCE 29
7.1 Environments and Deployment Strategy 29
7.2 Performance Optimisation 29
7.3 Security Features 30
7.4 Governance and Administration 30
CHAPTER 8 RESULTS AND DISCUSSION 31
8.1 Application Output 31
8.2 Achievement of Objectives 31
8.3 Comparison with the Existing System 32
8.4 Limitations of the Low-Code Approach 32
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum 32
CHAPTER 9 CONCLUSION AND FUTURE ENHANCEMENTS 33
9.1 Conclusion 33
9.2 Future Enhancements 33
9.3 Learning Outcomes and Career Relevance 33
REFERENCES 34
APPENDIX A ZOHO CREATOR FEATURE COVERAGE CHECKLIST 35
APPENDIX B COURSE OUTCOME MAPPING 37
APPENDIX C APPLICATION CONFIGURATION SUMMARY 38
C.1 Forms, Fields and References 38
C.2 Picklist Option Values 38
C.3 Dashboard, Profiles and Sharing Settings 39
APPENDIX D USER MANUAL 40
D.1 Accessing the Application 40
D.2 Instructions for the Administrator 40
D.3 Instructions for the Citizen 40
D.4 Instructions for the Police Officer 40
APPENDIX E EVALUATION RUBRIC 41
APPENDIX F VIVA-VOCE RECORD SHEET 43

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application
Table 2.2: Summary of Literature Survey
Table 3.1: Stakeholders and User Roles
Table 3.2: Functional Requirements
Table 3.3: Non-Functional Requirements
Table 3.4: Hardware and Software Requirements
Table 3.5: Use Case Description – Submit and Resolve a Crime Report
Table 4.1: Entities and Relationships
Table 4.2: Field Specification – Citizen Registration Form (Citizen_Registration)
Table 4.3: Field Specification – Crime Report Form Form (Crime_Report_Form)
Table 4.4: Field Specification – Police Officer Form (Police_Officer)
Table 4.5: Field Specification – Case Assignment Form (Case_Assignment)
Table 4.6: Field Specification – Investigation Details Form (Investigation_Details)
Table 4.7: Field Specification – Evidence Management Form (Evidence_Management)
Table 4.8: Field Specification – Case Resolution Form (Case_Resolution)
Table 4.9: Field Specification – Citizen Feedback Form (Citizen_Feedback)
Table 4.10: Role–Permission Matrix (as described in the project configuration)
Table 4.11: Sharing Rules Defined in the Application
Table 5.1: Reports Implemented
Table 5.2: Components of the Admin_Dashboard
Table 5.3: Device Support Configuration
Table 6.1: Test Cases
Table 6.2: Defect Log
Table 6.3: User Acceptance Feedback
Table 7.1: Deployment Checklist
Table 7.2: Performance Optimisation Techniques
Table 7.3: Security Controls Implemented
Table 8.1: Application Outputs and Where They Are Shown
Table 8.2: Mapping of Objectives to Outcomes
Table 8.3: Existing System versus Proposed Application
Table A.1: Feature Coverage Checklist
Table A.2: Mapping of Course Outcomes to Report Chapters
Table C.1: Summary of Forms
Table C.2: Picklist Fields and Their Options
Table A.3: Evaluation Rubric for the Individual Project
Table A.4: Performance Level Descriptors
Table A.5: Viva-Voce Questions and Responses
Table A.6: Marks Summary

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing System
Figure 3.1: Use Case Diagram of the Application
Figure 4.1: System Architecture of the Application
Figure 4.2: Entity–Relationship Diagram
Figure 4.3: Workflow / Process Flow of the Crime Report Life Cycle
Figure 4.4: Navigation Structure of the Crime Report Management Application
Figure 6.1: Debugging Cycle Used for Configuration Errors
Figure 7.1: Environments and Publish to Production (Promotion Flow)

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CO	Course Outcome
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FIR	First Information Report
FK / PK	Foreign Key / Primary Key
FR	Functional Requirement
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Crime reporting is one of the most important public services of a police department. When a citizen is a victim of theft, fraud, assault, cybercrime or any other offence, the first thing needed is a reliable way to record the complaint. After the complaint is recorded, many different activities follow: the complaint must be given to a suitable police officer, the officer must investigate and record findings, evidence must be collected and stored safely, the final outcome of the case must be approved and recorded, and the citizen should be able to tell the department how the complaint was handled.

In many police stations these activities are still handled with paper registers, separate files and verbal communication. Each step is maintained by a different person, and the information is rarely connected. A senior officer who wants to know how many theft complaints are open, which officer is handling a particular case, or whether evidence has been collected for a report has to search through registers and ask people. This slows down the work and makes it difficult to supervise.

The **Crime Report Management System** is a low-code application developed on **Zoho Creator** to bring all these activities into one online system. It records citizens, crime complaints, police officers, case assignments, investigations, evidence, case resolutions and citizen feedback in eight connected forms. Eight reports and one administrative dashboard give the users a live view of the information.

1.2 Problem Statement

Crime complaints and the work that follows them are not managed on one common platform. Complaint details, officer details, investigation notes and evidence information are kept in different places, so officers and administrators spend time searching for information, and citizens have no clear way to give feedback about the service. A single system is needed that connects every step of the crime report life cycle. The main pain points are:

- The same citizen, officer and complaint details are written again in several registers, which causes errors and duplicate data.
- There is no direct link between a complaint, the officer who handles it, the investigation done, the evidence collected and the final outcome.
- Supervisors cannot easily see how many complaints belong to each crime type or how many are submitted, under review, under investigation, resolved or closed.
- Evidence details and storage references are written in separate notes, so evidence can be lost or difficult to trace.

- Closure of a case is not recorded in a standard way, and there is no record of who approved the closure.
- Citizens have no simple method to rate the handling of their complaint or to give suggestions.
- Information is not protected by role, so records may be visible to people who do not need them.

1.3 Objectives

- To design a data model of eight connected forms for citizens, crime reports, police officers, case assignments, investigations, evidence, case resolutions and citizen feedback.
- To link the forms through picklist (lookup) references so that a crime report, an officer or a citizen is selected from existing records instead of being typed again.
- To capture the complaint location, including latitude and longitude, and to allow up to ten evidence files for a report.
- To provide eight reports, one for every form, so that users can view and manage the records.
- To build an Admin_Dashboard with donut and bar charts, an embedded case-assignment form and an embedded crime-report register for monitoring.
- To control access with profiles, roles and sharing rules so that administrators, police officers and citizens see only the records they need.
- To provide a section-based menu and to make the application usable on web, phone and tablet layouts.

1.4 Scope of the Project

In scope: registration of citizens; submission of crime complaints with type, date, time, location, description, status and evidence files; maintenance of police officer details; assignment of crime reports to officers with priority and instructions; recording of investigation details; management of evidence information; recording of case resolution and closure; collection of citizen feedback with a rating; eight reports; the Admin_Dashboard; profiles, roles and sharing rules; the Space navigation menu; and web, phone and tablet layouts.

Out of scope: integration with an official police or court database, automatic FIR number generation, SMS or e-mail alerts to citizens, automatic assignment of cases to officers, facial or fingerprint matching, payment or fine collection, and a public mobile application with push notifications. Some of these items are listed as future enhancements in Chapter 9.

1.5 Justification for a Low-Code Approach

A crime report system needs several forms, relationships between records, role-based security, reports and a dashboard. Building this with traditional programming would need a database design, a back-end, a front-end, hosting and a separate mobile application. Low-code platforms reduce this effort because the common parts are available as ready-made building blocks [4], [5]. Zoho Creator provides drag-and-drop form building, built-in reports and dashboards, role-based security, cloud hosting and mobile layouts in one place [1], [3]. This allows the application to be built quickly, changed easily when the department adds a new crime type or status, and maintained by a citizen developer. The cost of servers and maintenance is also avoided because Zoho hosts the application.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design, including the data model and access control. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing manual system, a citizen reports a crime by visiting the police station or by telephone. The duty officer writes the complaint in a paper register, and the details are later copied to files or spreadsheets. The case is given to an officer by a note or verbally. The officer keeps investigation notes in a case file, and evidence is kept in separate folders or lockers with handwritten labels. When the case ends, the closure is written in the register, and the citizen is seldom asked for feedback. The flow is shown below.

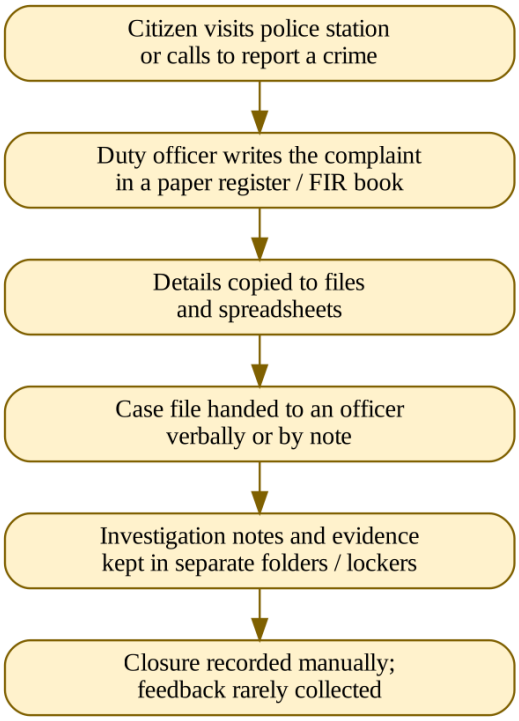


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Complaint, officer and citizen details are written more than once, so errors and mismatches are common.
- There is no direct link between the complaint, the assigned officer, the investigation, the evidence and the final result.
- Case status and crime-type statistics must be counted by hand, so management reports are slow and may be wrong.
- Evidence details and storage locations are not recorded in one place, which makes tracing evidence difficult.
- There is no standard record of who approved the closure of a case.

- Citizen feedback is not collected, so the quality of service cannot be measured.
- Anyone with access to the register or the spreadsheet can see all complaints, so data security is weak.

2.3 Low-Code versus Traditional Development

The table below compares the two approaches for this application. The effort values are estimates based on the size of this project (eight forms, eight reports, one dashboard, seven profiles and five sharing rules).

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	About 8–10 weeks (database, back-end, front-end, testing)	About 2–3 weeks (forms, reports and dashboard configured)
Skills required	Back-end, front-end and database developers	One citizen developer with basic Zoho Creator knowledge
Hosting and infrastructure	Own server or cloud set-up, backups and monitoring to be managed	Hosted and backed up by Zoho; nothing to install
Mobile application	Separate Android/iOS development needed	Phone and tablet layouts and the Zoho Creator mobile app
Security and access control	Login, roles and permissions coded manually	Built-in users, roles, profiles and sharing rules
Maintenance and change	Code change, testing and redeployment for each change	Form or report is edited visually and published
Customisation limits	Almost no limit	Limited by platform features; extended with Deluge when needed
Cost	High (developer time, servers, licences)	Low (platform subscription; trial and education editions available)

2.4 Literature Survey

The following sources were studied to understand low-code development and to decide the design of the application.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation, online documentation [1]	Zoho Creator Help Documentation	Describes forms, fields, reports, pages, user management and application settings.	Main reference for building every module.
2	Zoho Corporation, online documentation [2]	Deluge Script Reference Guide	Gives syntax for fetch, insert, validation and notifications.	Studied for future automation (Chapter 9).

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
3	A. C. Bock and U. Frank, 2021 [4]	Low-Code Platform	Defines low-code platforms and their main characteristics and limits.	Used to justify the approach and to discuss limits in Chapter 8.
4	R. Sanchis et al., 2020 [5]	Low-Code as Enabler of Digital Transformation in Manufacturing Industry	Shows that low-code speeds up digital transformation for small and medium organisations.	Similar benefit expected for a police administration office.
5	A. Sahay et al., 2020 [6]	Supporting the understanding and comparison of low-code development platforms	Compares platforms using common features such as data model, workflow and integration.	Helped to check that Zoho Creator covers the required features.

2.5 Emerging Trends in Low-Code Development

Several trends are shaping low-code development. Citizen development allows business users to build their own applications without waiting for a large IT team. AI-assisted building lets a developer describe an application in plain words and receive a first version of forms and reports, which can then be improved. Hyperautomation joins approvals, scripts, notifications and integrations into one automatic flow. Composable applications are built from small reusable parts that can be changed independently. The Crime Report Management System follows these ideas: every master and transaction record is a separate form, the forms are joined by picklist references, and the same design can be extended later with automation, notifications and AI features such as crime-pattern analysis.

2.6 Proposed System

The proposed system is the Crime Report Management System on Zoho Creator. It keeps all data in one cloud database and connects crime reports, police officers and citizens through picklist references, so details are selected from existing records instead of being typed again. A report receives an automatically generated Report_ID when a complaint is submitted. The report can then be assigned to an officer, investigated, supported with evidence, resolved and rated by the citizen. Eight reports show the records of every form, and the Admin_Dashboard shows crime types and case status with charts. Profiles, roles and sharing rules protect the data. In this way every limitation listed in Section 2.2 is addressed.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator / CEO	Owner of the application and its configuration	Manage users, profiles and sharing rules; maintain officer records; assign cases; view all records, reports and the Admin_Dashboard
Police Officer	Officer who handles complaints	View relevant crime reports and case assignments; record investigations; manage evidence; record case resolutions
Citizen	Member of the public	Register personal details; submit a crime report with evidence files; give feedback about the handling of a report

In the application, the roles **CEO**, **Police Officer** and **Citizen** are used in the sharing rules. The CEO role is used for administrative access, and the Administrator profile is configured with all permissions.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall store citizen details: name, date of birth, gender, mobile number, e-mail and address with location coordinates.	Citizen, Admin	High
FR-02	The system shall allow a complaint to be submitted with complainant details, crime type, incident date and time, location, description and up to 10 evidence files.	Citizen	High
FR-03	The system shall generate a unique Report_ID for every crime report and keep a report status (Submitted, Under Review, Under Investigation, Resolved, Closed).	System	High
FR-04	The system shall maintain police officer details: name, badge number, rank, police station and contact information.	Admin	High
FR-05	The system shall assign a crime report to a police officer with date, priority, instructions and assignment status.	Admin	High
FR-06	The system shall record investigation details: report, officer, date, findings, action taken, status and remarks.	Police Officer	High
FR-07	The system shall record evidence for a report: type, description, files, collected date, collecting officer, storage reference, remarks and status.	Police Officer	High
FR-08	The system shall record the case resolution: date, final outcome, summary, approving officer, closure status and remarks.	Police Officer	High

FR ID	Requirement	Role(s)	Priority
FR-09	The system shall collect citizen feedback with a rating from 1 to 5, comments, date and suggestions for a report.	Citizen	Medium
FR-10	The system shall provide a report for every form so that records can be viewed and managed.	All	High
FR-11	The system shall provide an Admin_Dashboard with a crime-type chart, two case-status charts, an embedded case-assignment form and an embedded crime-report register.	Admin	High
FR-12	The system shall control access through profiles, roles and sharing rules.	Admin	High
FR-13	The system shall provide a section-based menu and web, phone and tablet layouts.	All	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard shall load within a few seconds for the expected number of records.
NFR-02	Security	Users shall access only the forms, reports and records permitted by their profile, role and the sharing rules.
NFR-03	Usability	The application shall be usable on a desktop browser, a phone and a tablet with suitable layouts and clear labels.
NFR-04	Availability	The application shall be available whenever the Zoho Creator cloud service is available (hosted by Zoho).
NFR-05	Maintainability	Forms and fields shall follow a clear naming convention (words joined by underscores) so that another developer can change them easily.
NFR-06	Data integrity	Related records shall be selected through picklist references so that the same report, officer or citizen is not typed again.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Trial / Education / Professional Edition)
Application type	Low-code, configuration-based application (forms, reports, dashboard, access control)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Mobile	Phone and tablet layouts in a mobile browser or the Zoho Creator mobile app
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection (or any smartphone or tablet with internet)

3.5 Use Case Model

The use case diagram shows the three user roles and the main actions each role performs in the application.

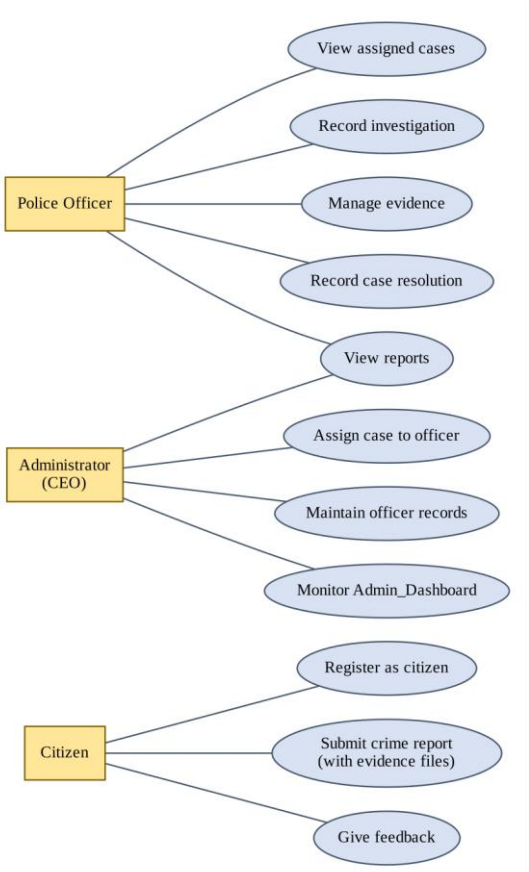


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Submit and Resolve a Crime Report

Item	Description
Use Case ID	UC-01
Use Case Name	Submit and Resolve a Crime Report
Actor(s)	Citizen, Administrator, Police Officer
Pre-condition	The citizen is registered and the police officer records exist in the system.
Main Flow	1. The citizen opens the Crime Report Form and enters the complainant details, crime type, incident date and time, location and description, and uploads evidence files. 2. The system generates the Report_ID and the report appears in the Crime Report Form Report. 3. The administrator opens the Case Assignment form (also embedded in the dashboard), selects the Report_ID and the officer, and sets the date, priority and instructions. 4. The officer records the investigation findings and the action taken in Investigation Details and stores evidence in Evidence Management. 5. The officer records the final outcome and closure in Case Resolution. 6. The citizen gives a rating and comments in Citizen Feedback.
Alternate Flow	If the evidence is not enough, the Final_Outcome is recorded as Insufficient Evidence. If the case is passed to another authority, the outcome is Case Referred. If the closure is not accepted, the Closure value is Rejected.
Post-condition	The crime report carries its latest status, and linked assignment, investigation, evidence, resolution and feedback records can be traced through the Report_ID.
Related FR	FR-02 to FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The application follows the standard architecture of a Zoho Creator application. Users reach the application through a web browser, a phone or a tablet. The **presentation layer** consists of the Space menu, eight forms, eight reports and the Admin_Dashboard. The **access control layer** (profiles, roles and sharing rules) decides what each user can open and see. The **data layer** is the Zoho Creator cloud database, in which the records of the eight forms are connected through picklist references. Zoho hosts and maintains all layers, so no server has to be managed by the developer.

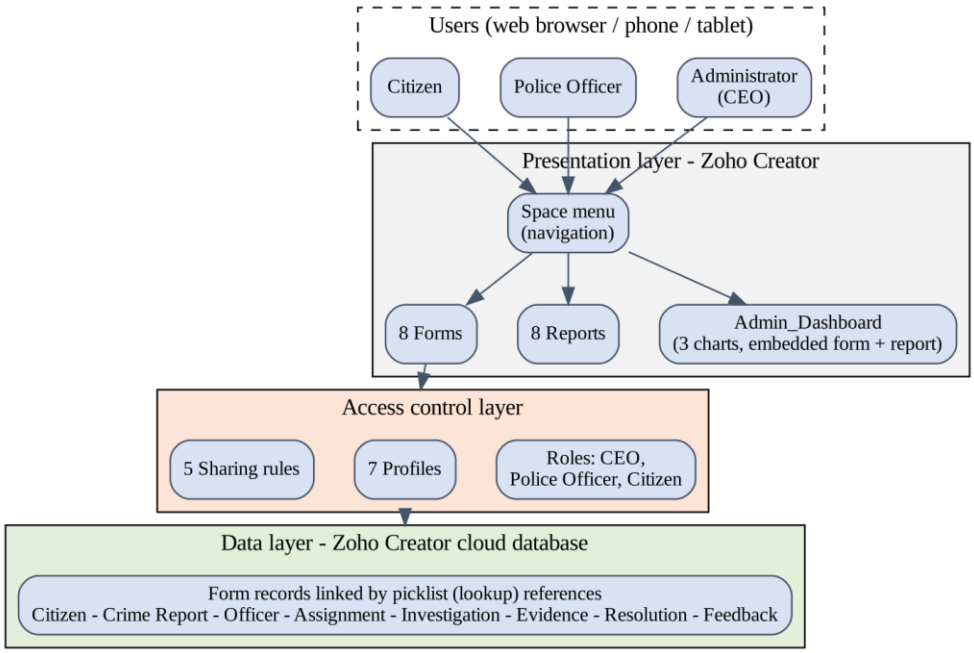


Figure 4.1: System Architecture of the Application

4.2 Data Model

The data model has eight forms. Three forms store master information (Citizen_Registration, Police_Officer and Crime_Report_Form), and five forms store transactions that refer to them (Case_Assignment, Investigation_Details, Evidence_Management, Case_Resolution and Citizen_Feedback). Every form has an auto-number identifier. The relationships are created with **picklist fields that reference records of another form**; in the application, the picklist stores the selected record and shows a readable value such as Report_ID, Officer_Name or Full_Name.

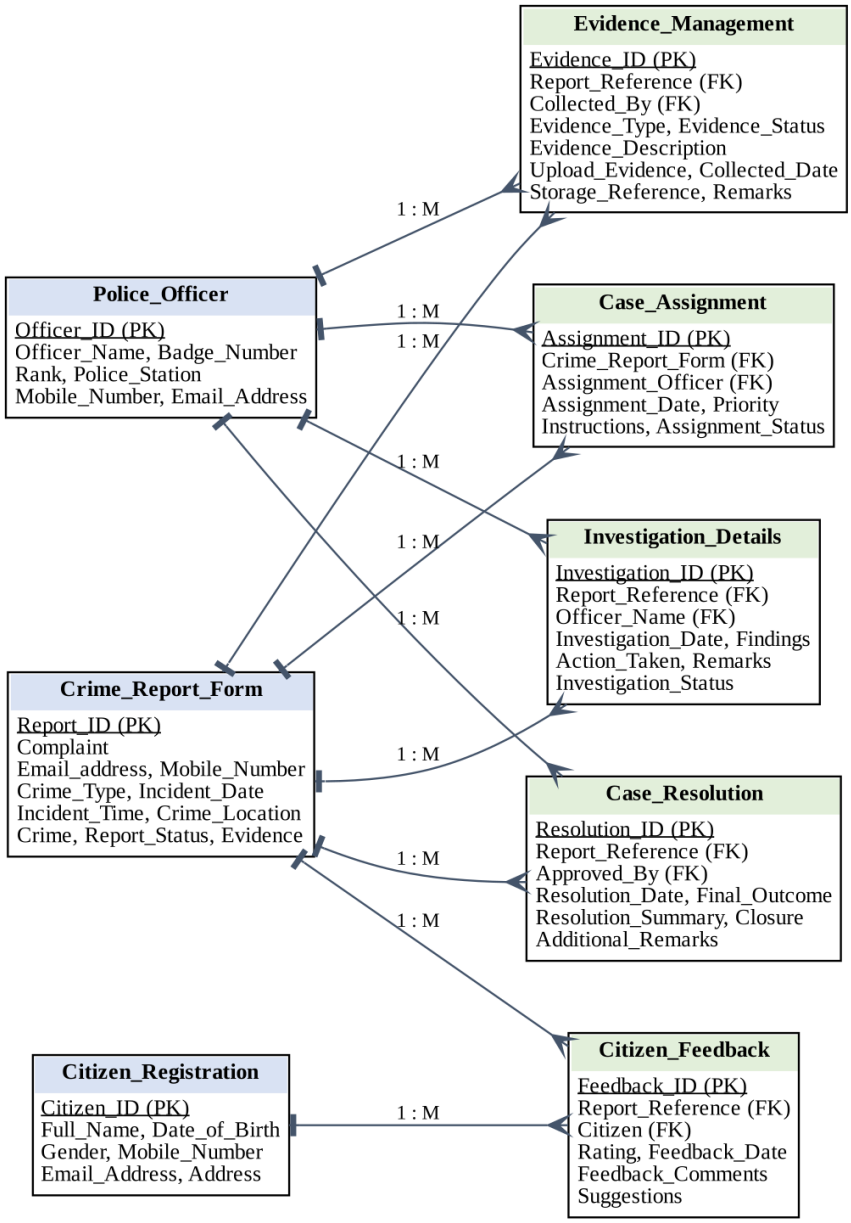


Figure 4.2: Entity–Relationship Diagram

A crime report can be referenced by many assignment, investigation, evidence, resolution and feedback records. A police officer can be referenced by many assignments, investigations, evidence records and resolutions. A citizen can be referenced by many feedback records. All relationships are therefore one-to-many from the master form to the transaction form. Table 4.1 lists every relationship.

Table 4.1: Entities and Relationships

S.No.	Form (Many side)	Picklist Field	References (One side)	Displayed Value
1	Case_Assignment	Crime_Report_Form	Crime_Report_Form	Report_ID
2	Case_Assignment	Assignment_Officer	Police_Officer	Officer_Name

S.No.	Form (Many side)	Picklist Field	References (One side)	Displayed Value
3	Investigation_Details	Report_Reference	Crime_Report_Form	Report_ID
4	Investigation_Details	Officer_Name	Police_Officer	Officer_Name
5	Evidence_Management	Report_Reference	Crime_Report_Form	Report_ID
6	Evidence_Management	Collected_By	Police_Officer	Officer_Name
7	Case_Resolution	Report_Reference	Crime_Report_Form	Report_ID
8	Case_Resolution	Approved_By	Police_Officer	Officer_Name
9	Citizen_Feedback	Report_Reference	Crime_Report_Form	Report_ID
10	Citizen_Feedback	Citizen	Citizen_Registration	Full_Name

Design observation. Crime_Report_Form stores the complainant as text (Complaint, Email_address and Mobile_Number) and is not linked to Citizen_Registration. A citizen is connected to a report only through Citizen_Feedback. This keeps the complaint form simple, so that a person can report a crime quickly, and a direct link between a report and a registered citizen is listed as a future enhancement in Chapter 9.

4.3 Form Design

The eight forms are described below. For every form the table shows the field label, the link name used inside Zoho Creator, the field type and the options or remarks. In total the forms contain 65 fields of 10 different field types: Auto Number, Text, Date, Time, Picklist, Phone Number, Email, Address, Text Area and File Upload.

Table 4.2: Field Specification – Citizen Registration Form (Citizen_Registration)

Field Label	Link Name	Field Type	Options / Remarks
Citizen ID	Citizen_ID	Auto Number	Unique number generated by the system
Full Name	Full_Name	Text	Name of the citizen; shown in the Citizen picklist of Citizen_Feedback
Date of Birth	Date_of_Birth	Date	Date of birth of the citizen
Gender	Gender	Picklist	Male / Female / Transgender
Mobile Number	Mobile_Number	Phone Number	Contact number
Email Address	Email_Address	Email	Contact e-mail
Address	Address	Address	Address Line 1, Address Line 2, City/District, State/Province, Postal Code, Country, Latitude, Longitude. Latitude and Longitude are hidden; coordinate capture is enabled

Table 4.3: Field Specification – Crime Report Form Form (Crime_Report_Form)

Field Label	Link Name	Field Type	Options / Remarks
Report ID	Report_ID	Auto Number	Unique report number; displayed in all report picklists
Complaint (Complainant Name)	Complaint	Text	Name of the person making the complaint
Email address	Email_address	Email	Complainant e-mail
Mobile Number	Mobile_Number	Phone Number	Complainant mobile number
Crime Type	Crime_Type	Picklist	Theft / Robbery / Cybercrime / Assault / Missing Person / Fraud / Property Damage / Other
Incident Date	Incident_Date	Date	Date on which the incident happened
Incident Time	Incident_Time	Time	Time of the incident
Crime Location	Crime_Location	Address	Address lines, city/district, state/province, postal code, country, latitude, longitude; coordinate capture enabled
Crime (Crime Description)	Crime	Text Area	Description of the crime
Report Status	Report_Status	Picklist	Submitted / Under Review / Under Investigation / Resolved / Closed
Evidence	Evidence	File Upload	Maximum 10 files

Table 4.4: Field Specification – Police Officer Form (Police_Officer)

Field Label	Link Name	Field Type	Options / Remarks
Officer ID	Officer_ID	Auto Number	Unique number generated by the system
Officer Name	Officer_Name	Text	Displayed in all officer picklists
Badge Number	Badge_Number	Text	Official badge number
Rank	Rank	Picklist	Inspector / Sub-Inspector / Head Constable / Constable / Other
Police Station	Police_Station	Text	Station where the officer works
Mobile Number	Mobile_Number	Phone Number	Contact number
Email Address	Email_Address	Email	Contact e-mail

Table 4.5: Field Specification – Case Assignment Form (Case_Assignment)

Field Label	Link Name	Field Type	Options / Remarks
Assignment ID	Assignment_ID	Auto Number	Unique number generated by the system

Field Label	Link Name	Field Type	Options / Remarks
Crime Report Form	Crime_Report_Form	Picklist (reference)	References Crime_Report_Form; displays Report_ID
Assignment Officer	Assignment_Officer	Picklist (reference)	References Police_Officer; displays Officer_Name
Assignment Date	Assignment_Date	Date	Date of assignment
Priority	Priority	Picklist	Low / Medium / High
Instructions	Instructions	Text Area	Instructions given to the officer
Assignment Status	Assignment_Status	Picklist	Assigned / In progress / Completed

Table 4.6: Field Specification – Investigation Details Form (Investigation_Details)

Field Label	Link Name	Field Type	Options / Remarks
Investigation ID	Investigation_ID	Auto Number	Unique number generated by the system
Report Reference	Report_Reference	Picklist (reference)	References Crime_Report_Form; displays Report_ID
Officer Name	Officer_Name	Picklist (reference)	References Police_Officer; displays Officer_Name
Investigation Date	Investigation_Date	Date	Date of the investigation
Findings	Findings	Text Area	Findings of the investigation
Action Taken	Action_Taken	Text Area	Action taken by the officer
Investigation Status	Investigation_Status	Picklist	Not Started / In Progress / Evidence Collected / Completed
Remarks	Remarks	Text Area	Additional remarks

Table 4.7: Field Specification – Evidence Management Form (Evidence_Management)

Field Label	Link Name	Field Type	Options / Remarks
Evidence ID	Evidence_ID	Auto Number	Unique number generated by the system
Report Reference	Report_Reference	Picklist (reference)	References Crime_Report_Form; displays Report_ID
Evidence Type	Evidence_Type	Picklist	Photograph / Video / Document / Fingerprint / Physical Evidence / Digital Evidence / Other
Evidence Description	Evidence_Description	Text Area	Description of the evidence
Upload Evidence	Upload_Evidence	File Upload	Maximum 10 files
Collected Date	Collected_Date	Date	Date on which evidence was collected
Collected By	Collected_By	Picklist (reference)	References Police_Officer; displays Officer_Name

Field Label	Link Name	Field Type	Options / Remarks
Storage Reference	Storage_Reference	Text	Place or reference where physical evidence is stored
Remarks	Remarks	Text Area	Additional remarks
Evidence Status	Evidence_Status	Picklist	Collected / Under review / Verified

Table 4.8: Field Specification – Case Resolution Form (Case_Resolution)

Field Label	Link Name	Field Type	Options / Remarks
Resolution ID	Resolution_ID	Auto Number	Unique number generated by the system
Report Reference	Report_Reference	Picklist (reference)	References Crime_Report_Form; displays Report_ID
Resolution Date	Resolution_Date	Date	Date on which the case was resolved
Final Outcome	Final_Outcome	Picklist	Suspect Arrested / Property Recovered / Complaint Resolved / Insufficient Evidence / Case Referred / Other
Resolution Summary	Resolution_Summary	Text Area	Summary of the resolution
Approved By	Approved_By	Picklist (reference)	References Police_Officer; displays Officer_Name
Closure	Closure	Picklist	Pending Approval / Approved / Rejected / Closed
Additional Remarks	Additional_Remarks	Text Area	Additional remarks

Table 4.9: Field Specification – Citizen Feedback Form (Citizen_Feedback)

Field Label	Link Name	Field Type	Options / Remarks
Feedback ID	Feedback_ID	Auto Number	Unique number generated by the system
Report Reference	Report_Reference	Picklist (reference)	References Crime_Report_Form; displays Report_ID
Citizen	Citizen	Picklist (reference)	References Citizen_Registration; displays Full_Name
Rating	Rating	Picklist	1 – Very Poor / 2 – Poor / 3 – Average / 4 – Good / 5 – Excellent
Feedback Comments	Feedback_Comments	Text Area	Comments of the citizen
Feedback Date	Feedback_Date	Date	Date of the feedback
Suggestions	Suggestions	Text Area	Suggestions to improve the service

4.4 Process and Workflow Design

The application follows the life cycle of a crime report. The citizen information and the complaint are entered first; the Report_ID is generated by the system; the report is assigned to an officer; the investigation and evidence are recorded; the case is resolved and closed; and the citizen gives feedback. The administrator monitors the whole flow through the reports and the Admin_Dashboard. The forms are connected through the picklist references explained in Section 4.2, so the flow is carried by the data model.

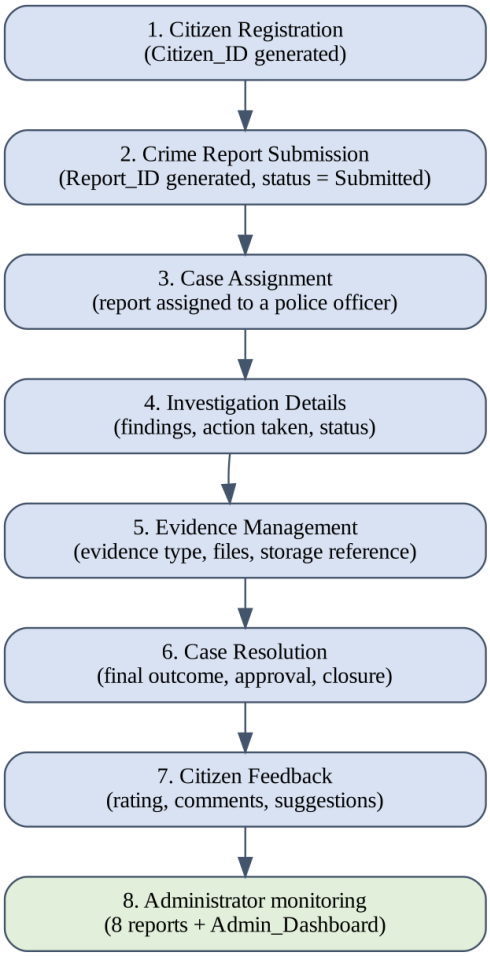


Figure 4.3: Workflow / Process Flow of the Crime Report Life Cycle

Report status values (Submitted, Under Review, Under Investigation, Resolved, Closed), assignment status values (Assigned, In progress, Completed), investigation status values (Not Started, In Progress, Evidence Collected, Completed), evidence status values (Collected, Under review, Verified) and closure values (Pending Approval, Approved, Rejected, Closed) are all picklists, so the progress of a case is recorded in a standard way at each stage.

4.5 User Interface and Navigation Design

The application menu is named **Space**. It is arranged in sections, one for every form together with its report, so that a user finds the form and the matching report next to each other. Other sections are App Preferences, Approvals (with Pending Tasks and Completed Tasks), Reports and the Admin Dashboard. The structure is shown below.

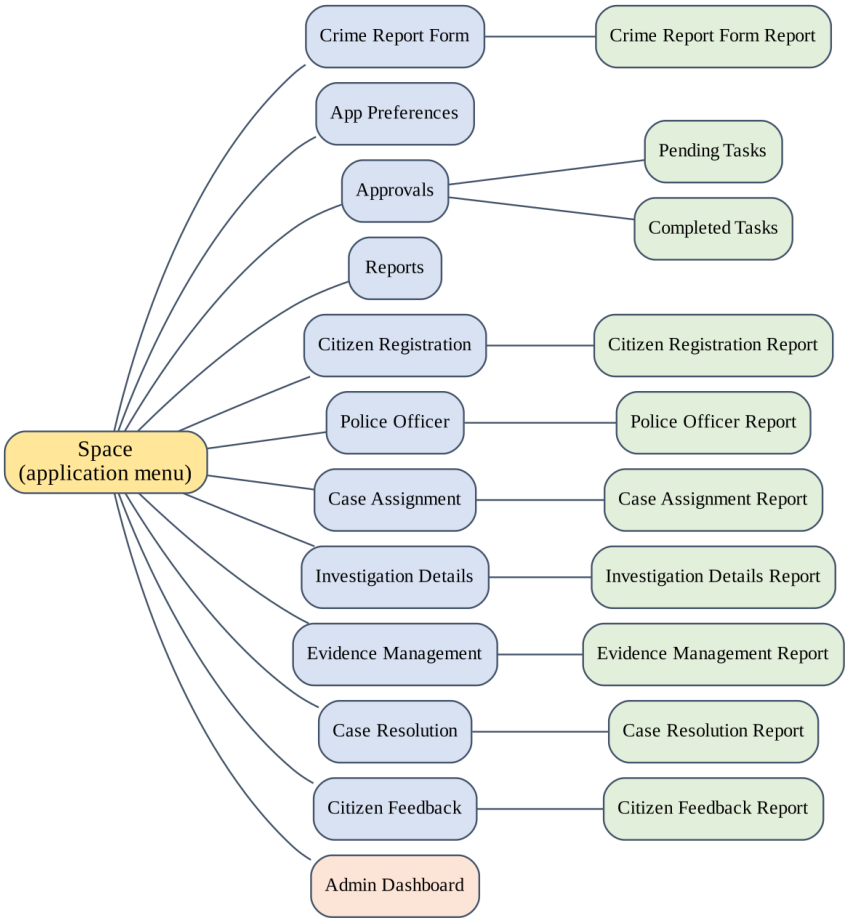


Figure 4.4: Navigation Structure of the Crime Report Management Application

The user interface supports three layouts. The **web** layout uses the Lato font, left-aligned form labels and section-based navigation, and the application icon text is CRM. The **phone** layout uses automatic label placement, a two-column layout and the Chrysler Purple colour option. The **tablet** layout uses automatic label placement, a sliding-pane layout and the Dark Grey colour option. All eight forms have both phone and tablet layouts.

4.6 Access Control Design

Access control has three parts. **Profiles** define what a user may do (permissions). **Roles** define where a user stands in the organisation hierarchy and which records the user can see. **Sharing rules** give users of one role access to records created by another role. The application contains the profiles Administrator, Police_Officer_Permissions, Citizen_Permissions, Read, Write, Developer and Customer; the Administrator profile is

configured with all permissions. The user roles used in the sharing rules are CEO, Police Officer and Citizen.

Table 4.10: Role–Permission Matrix (as described in the project configuration)

Module / Form	Administrator (CEO)	Police Officer	Citizen
Citizen Registration	Full access	—	Add own details
Crime Report Form	Full access	Access to relevant reports (Read via sharing from Citizen)	Submit report
Police Officer	Full access	—	—
Case Assignment	Full access (also from dashboard)	Access to relevant assignments (Read via sharing from CEO)	—
Investigation Details	Full access	Access to investigation records (Read via sharing from CEO)	—
Evidence Management	Full access	Access to evidence records (Read via sharing from CEO)	—
Case Resolution	Read (via sharing from Police Officer)	Access to case resolution; shared to CEO	—
Citizen Feedback	Full access	—	Submit feedback
Admin_Dashboard	Full access	—	—

"—" means that the access is not described in the application configuration. Administrator access is complete because the Administrator profile has all permissions.

The application defines five sharing rules. All of them give **Read** access, so the receiving role can see the records but cannot change them.

Table 4.11: Sharing Rules Defined in the Application

S.No.	Sharing Rule	Form Shared	Shared From	Shared To	Access
1	Share_Investigation_Details_With_Police	Investigation Details	CEO	Police Officer	Read
2	Share_Evidence_With_Police	Evidence Management	CEO	Police Officer	Read
3	Share_Case_Assignments_With_Police	Case Assignment	CEO	Police Officer	Read
4	Share_Crime_Reports_With_Police	Crime Report Form	Citizen	Police Officer	Read
5	Share_Case_Resolutions_With_Admin	Case Resolution	Police Officer	CEO	Read

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application **Crime_report_management_system** was created in Zoho Creator from scratch. The application menu, named **Space**, was arranged in sections. The web settings use the Lato font with left-aligned form labels, and the application icon text is CRM. The application contains eight forms, eight reports and one dashboard (Admin_Dashboard).

5.2 Forms and Data Capture

Each form was created in the Zoho Creator form builder, and the fields were added by drag and drop. Auto Number fields generate the identifiers, Picklist fields restrict entries to standard values, and picklist reference fields select records of another form. The special features used are described below.

- **Address fields with coordinate capture:** the Address field of Citizen_Registration and the Crime_Location field of Crime_Report_Form contain address lines, city/district, state/province, postal code, country, latitude and longitude. Coordinate capture is enabled, and in Citizen_Registration the latitude and longitude fields are hidden from the user.
- **File upload:** Crime_Report_Form (Evidence) and Evidence_Management (Upload_Evidence) allow a maximum of 10 files each.
- **Date and time fields:** Incident_Date and Incident_Time record when the crime happened; other forms record assignment, investigation, collection, resolution and feedback dates.
- **Picklist references:** Case_Assignment, Investigation_Details, Evidence_Management, Case_Resolution and Citizen_Feedback select the report (Report_ID), the officer (Officer_Name) or the citizen (Full_Name) from existing records.
- **Embedded form:** the Case_Assignment form is embedded in the dashboard with the title “Assign Crime Case”, the submit label “Assign Case” and the success message “Case assigned successfully!”.

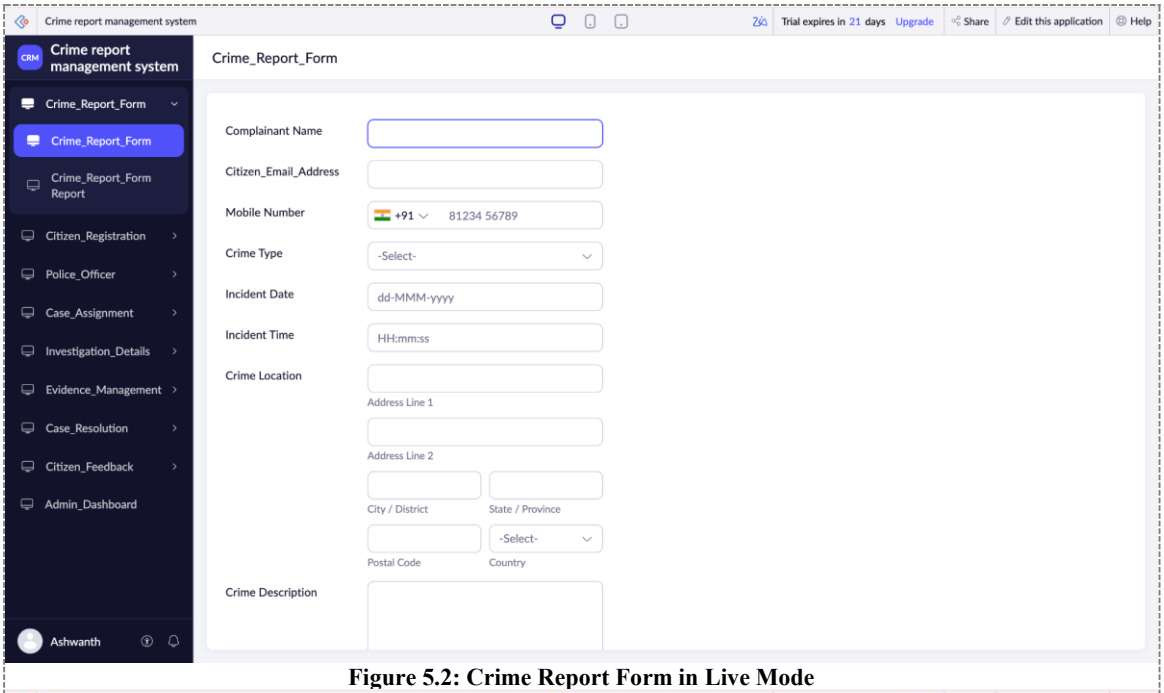
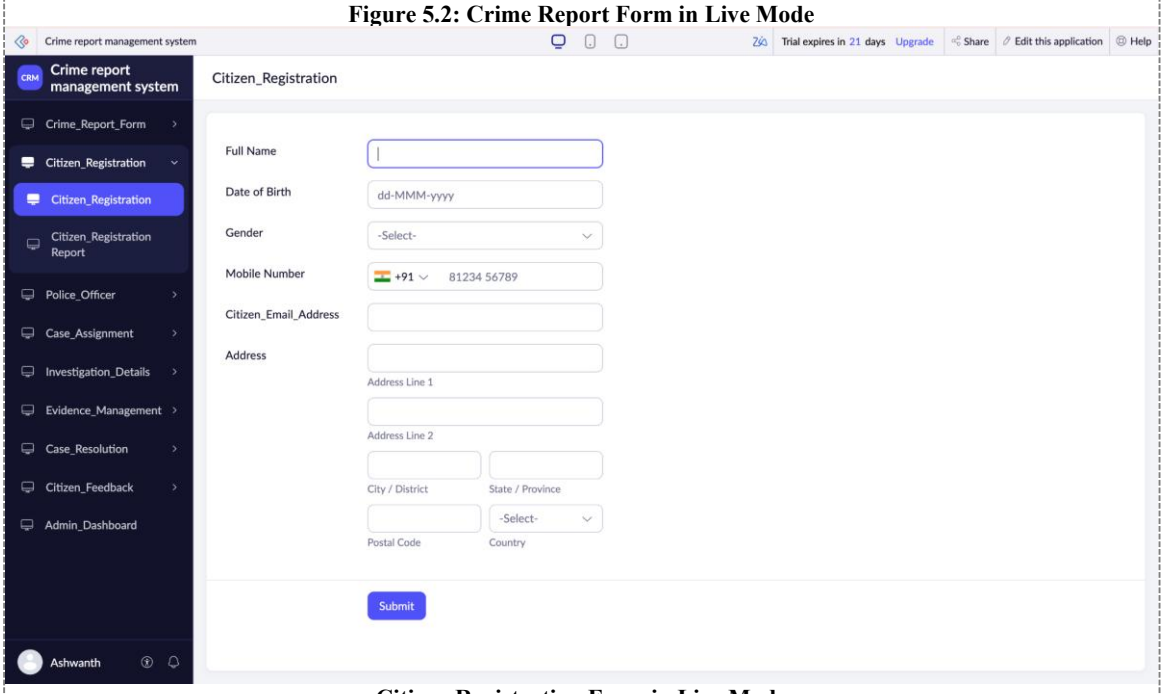


Figure 5.2: Crime Report Form in Live Mode



Citizen Registration Form in Live Mode

5.3 Reports

One report was created for every form. Reports show the records in a table, allow users to open, search and sort records, and can be used to edit or delete records according to the user's permissions.

Table 5.1: Reports Implemented

S.No.	Report Name	Based on Form	Purpose
1	Crime_Report_Form_Report	Crime_Report_Form	Lists all crime complaints with type, date, location and status; used to track complaints

S.No.	Report Name	Based on Form	Purpose
2	Citizen_Registration_Report	Citizen_Registration	Lists registered citizens and their contact information
3	Police_Officer_Report	Police_Officer	Lists police officers with badge number, rank and station
4	Case_Assignment_Report	Case_Assignment	Shows which report is assigned to which officer, with priority and status
5	Investigation_Details_Report	Investigation_Details	Shows investigation findings, actions and status for each report
6	Evidence_Management_ Report	Evidence_ Management	Shows evidence type, collecting officer, storage reference and status
7	Case_Resolution_Report	Case_Resolution	Shows final outcome, approving officer and closure status
8	Citizen_Feedback_Report	Citizen_Feedback	Shows ratings, comments and suggestions given by citizens

5.4 Pages and Dashboards

The **Admin_Dashboard** was built with the Zoho Creator dashboard builder. It contains three charts and two embedded components that give the administrator a single screen for monitoring.

Table 5.2: Components of the Admin_Dashboard

S.No.	Component	Type	Source / Configuration	Purpose
1	Crime Reports by Type	Donut chart	Category: Crime_Type; Value: Count of ID	Shows the distribution of complaints by crime category
2	Cases by Status	Donut chart	Category: Report_Status; Value: Count of ID	Shows the share of cases in each status
3	Cases by Status	Bar chart	Category: Report_Status; Value: Count of ID	Compares the number of cases in different statuses
4	Assign Crime Case	Embedded form	Case_Assignment form; submit label “Assign Case”; message “Case assigned successfully!”	Lets the administrator assign a case without leaving the dashboard
5	Crime Reports Register	Embedded report	Crime_Report_Form_Report; bulk edit, bulk delete, bulk duplicate and duplicate-record controls disabled	Displays crime report records and statuses safely

The charts are updated whenever a report is added or its status changes. The administrator can see at a glance which crime types are most common and how many cases are submitted, under review, under investigation, resolved or closed. The embedded form allows quick assignment of a new report, and the embedded register shows the latest records. Disabling bulk edit, bulk delete and duplicate options in the embedded report protects the data from accidental changes made from the dashboard.

5.5 Scripting and Workflow Automation

The Crime Report Management System is built **entirely with configuration**: forms, fields, picklist references, reports, the dashboard, profiles, roles and sharing rules. No custom Deluge scripts, form workflows, schedules or approval processes are defined in the application. The identifiers are generated by Auto Number fields, the values are restricted by picklists, and the dashboard charts calculate counts automatically. This shows the strength of the low-code approach: a complete working system was produced without writing code. Deluge automation, such as sending an e-mail when a report is assigned or marking a report as Under Investigation when an investigation record is added, is planned as a future enhancement (Chapter 9).

5.6 Integrations and Built-in Platform Features

The application does not integrate with any external system. It uses the following built-in platform features: address fields with location coordinate capture, file upload fields for evidence, the Space menu with an Approvals section (Pending Tasks and Completed Tasks), and the cloud hosting, backup and security provided by Zoho. The Approvals section is part of the application menu; no custom approval process is configured.

5.7 User Management and Role-Based Access

Users are added in Zoho Creator and assigned a role and a profile. The Administrator profile has all permissions. The Police_Officer_Permissions profile gives police officers access to relevant crime reports, case assignments, investigation records, evidence management and case resolution. The Citizen_Permissions profile gives citizens access to registration, crime report submission and feedback. The profiles Read, Write, Developer and Customer are available for other kinds of users. The five sharing rules in Table 4.11 give Read access to records across roles, for example crime reports from Citizen to Police Officer and case resolutions from Police Officer to CEO.

5.8 AI and Analytics Features

The analytics of the application are provided by the dashboard charts (two donut charts and one bar chart) that count crime reports by type and by status. Zia, the AI assistant of Zoho, was not used for this version. AI features such as crime-pattern prediction, duplicate-complaint detection and automatic summarisation of investigation notes are suggested as future enhancements.

5.9 Pro-Code Extension

No pro-code extension (custom functions, widgets or Zoho Catalyst) was used in this version. The application can be extended later using Deluge custom functions and the Zoho Creator REST API, for example to send SMS alerts or to connect with an external records system.

5.10 Mobile and Tablet Access

Phone and tablet layouts were created for all eight forms. The phone layout uses automatic label placement, a two-column layout and the Chrysler Purple colour option, so that the fields are easy to read on a small screen. The tablet layout uses automatic label placement, a sliding-pane layout and the Dark Grey colour option. These layouts are valuable because citizens usually report crimes from a phone, officers collect evidence and update records in the field using a phone or tablet, and administrators prefer the web layout for the dashboard.

Table 5.3: Device Support Configuration

Device	Forms Covered	Label Placement	Layout	Colour / Font / Other
Web	All forms	Left-aligned labels	Section-based navigation	Lato font; icon text CRM
Phone	All 8 forms	Automatic	Two-column layout	Chrysler Purple
Tablet	All 8 forms	Automatic	Sliding-pane layout	Dark Grey

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Because the application is built with configuration instead of custom scripts, testing concentrates on the behaviour of forms, picklist references, reports, the dashboard, access control and device layouts. Testing was planned at four levels:

- **Form testing:** each form is opened in Access mode, filled with valid sample data and submitted; field types, picklist options, auto numbers, file limits and address coordinates are checked.
- **Relationship testing:** records are added in the transaction forms and the picklists are checked to confirm that they show the correct Report_ID, Officer_Name or Full_Name.
- **Report and dashboard testing:** the eight reports and the dashboard charts are compared with the records entered, and the embedded form and report are checked.
- **Access and device testing:** the application is opened with different users (Administrator, Police Officer, Citizen) to check profiles and sharing rules, and with phone and tablet preview to check the layouts.

6.2 Test Cases

The test cases below were prepared from the requirements in Chapter 3. The last column is filled with Pass or Fail when the test is executed in the live application.

Table 6.1: Test Cases

TC ID	Module	Test Description	Expected Result	Result
TC-01	Citizen Registration	Submit the form with name, date of birth, gender, mobile, e-mail and address.	Record is saved and Citizen_ID is generated automatically.	
TC-02	Citizen Registration	Enter an address and check the latitude and longitude fields.	Coordinates are captured; the latitude and longitude fields are hidden from the user.	
TC-03	Crime Report Form	Submit a complaint with crime type, date, time, location and description.	Report is saved and Report_ID is generated automatically.	
TC-04	Crime Report Form	Open the Crime_Type and Report_Status picklists.	Eight crime types and five statuses are listed.	
TC-05	Crime Report Form	Upload 10 evidence files, then try an 11th file.	10 files are accepted; the 11th file is not accepted.	
TC-06	Police Officer	Add an officer and choose a rank.	Record is saved; the Rank picklist shows five options.	

TC ID	Module	Test Description	Expected Result	Result
TC-07	Case Assignment	Open the picklists for report and officer.	Report_ID and Officer_Name values of existing records are displayed.	
TC-08	Case Assignment (dashboard)	Submit the embedded “Assign Crime Case” form.	Submit button reads “Assign Case”; message “Case assigned successfully!” is shown.	
TC-09	Investigation Details	Add an investigation for an existing report and officer.	Record is saved; Investigation_ Status shows four options.	
TC-10	Evidence Management	Add evidence with type, files, collecting officer and storage reference.	Record is saved and linked to the selected report and officer.	
TC-11	Case Resolution	Add a resolution with final outcome, approver and closure.	Record is saved; Final_ Outcome has six options and Closure has four.	
TC-12	Citizen Feedback	Select a report and citizen, then choose a rating.	Citizen picklist shows Full_ Name; Rating shows 1 – Very Poor to 5 – Excellent.	
TC-13	Reports	Open each of the eight reports.	Records of the matching form are displayed.	
TC-14	Dashboard	Add reports of different crime types and compare with the donut chart.	Crime Reports by Type chart shows the correct count for each type.	
TC-15	Dashboard	Change a Report_Status and refresh the dashboard.	Both Cases by Status charts show the updated counts.	
TC-16	Dashboard	Select records in the embedded Crime Reports Register.	Bulk edit, bulk delete, bulk duplicate and duplicate-record options are not available.	
TC-17	Access control	Sign in as a police officer and open crime reports, assignments, investigations and evidence.	Records are visible through the sharing rules with Read access.	
TC-18	Access control	Sign in as CEO and open Case Resolution records created by a police officer.	Resolution records are visible with Read access.	
TC-19	Phone layout	Open a form in the phone preview.	Two-column layout with automatic label placement is displayed.	
TC-20	Tablet layout	Open a form in the tablet preview.	Sliding-pane layout with automatic label placement is displayed.	

6.3 Debugging Approach

When a test case does not give the expected result, the cause is located by following the debugging cycle in Figure 6.1. In a configuration-based application, the usual causes are a wrong field type, a missing picklist option, a picklist reference that points to the wrong form or display field, a report that does not show a column, or a profile or sharing rule that

does not allow access. The cause is corrected in the form builder, report settings or access settings, and the test case is repeated. If Deluge scripts are added in future, info statements and the execution log of Zoho Creator will be used to follow the script step by step.

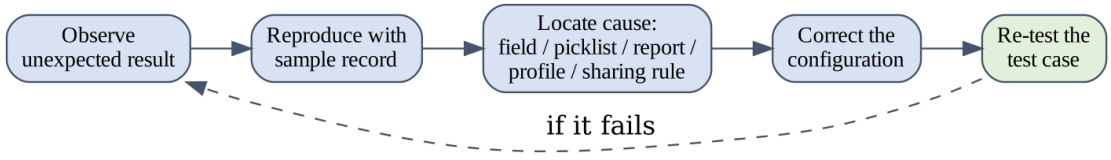


Figure 6.1: Debugging Cycle Used for Configuration Errors

6.4 Defect Log

Defects found during testing are recorded in the table below with their cause and correction. The table is filled in during the execution of the test cases.

Table 6.2: Defect Log

Defect ID	Module / Test Case	Description	Cause	Correction	Status
D-01					
D-02					
D-03					
D-04					

6.5 User Acceptance Testing

After internal testing, one user from each category tries the application and gives feedback. The tasks listed below are performed and the feedback is entered in the table.

Table 6.3: User Acceptance Feedback

User Role	Tasks Performed	Feedback	Accepted (Y/N)
Citizen	Register, submit a crime report with evidence, give feedback		
Police Officer	View assigned cases, record investigation and evidence, record resolution		
Administrator	Assign a case, view reports, use the Admin_Dashboard		

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application is built and tested in the **Development** environment with sample data. After the test cases pass and the user acceptance feedback is accepted, it is published to **Production** with version notes. In production, users, profiles and roles are configured again, and the sample data is removed. If a serious error is found after release, the previous version is restored using the version history, and the fix is made again in development.

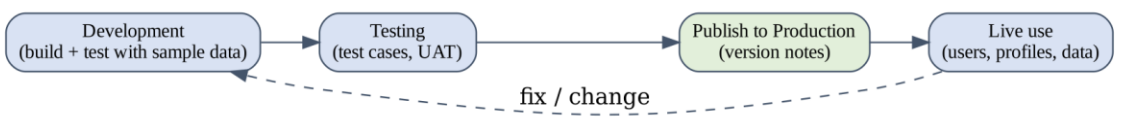


Figure 7.1: Environments and Publish to Production (Promotion Flow)

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development	
2	Sample / test data removed or isolated	
3	Users assigned to roles and profiles in Production	
4	Sharing rules checked with Administrator, Police Officer and Citizen users	
5	Phone and tablet layouts verified	
6	Application published and URL verified	
7	Version / release notes recorded	

7.2 Performance Optimisation

The following design choices keep the application light and fast. The effect was not measured with a timer in this version; measured values can be added after load testing.

Table 7.2: Performance Optimisation Techniques

Technique	Component	Benefit
Picklist references instead of repeated text	Five transaction forms	Report, officer and citizen details are stored once and referenced, which reduces data size
Auto Number identifiers	All eight forms	Identifiers are generated by the platform without extra processing
Aggregated chart values (Count of ID)	Admin_Dashboard charts	Charts show counts without loading every record
One report per form	Eight reports	Each report loads only the records of one form

Technique	Component	Benefit
Bulk operations disabled in embedded report	Crime Reports Register	Avoids heavy bulk actions from the dashboard
File limit of 10 per field	Evidence and Upload_ Evidence	Controls storage use and upload time
No custom scripts	Whole application	No script execution time; fewer API and statement limits to manage

7.3 Security Features

Security is applied at several levels. Profile permissions control which forms, reports and the dashboard a user can open and edit. Roles and sharing rules control which records a user can see; all five sharing rules are Read-only. The latitude and longitude fields of the citizen address are hidden from the user. Bulk edit and bulk delete are disabled in the embedded dashboard report. Zoho Creator serves the application over a secure (HTTPS) connection and hosts it on Zoho's cloud, and two-factor authentication of the Zoho account is recommended for administrators. Because the application stores personal and crime information, access should be given only to authorised persons.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Profiles (Administrator, Police_ Officer_Permissions, Citizen_ Permissions, Read, Write, Developer, Customer)	All forms, reports and the dashboard	Unauthorised access to data and functions
Roles (CEO, Police Officer, Citizen)	Data visibility	Users seeing records of other roles
Five Read-only sharing rules	Crime reports, assignments, investigations, evidence, resolutions	Unwanted change of shared records
Hidden latitude and longitude fields	Citizen_Registration address	Exposure of exact coordinates on the form
Disabled bulk actions	Embedded Crime Reports Register	Accidental mass edit or deletion
Picklist-restricted values	Status, type and outcome fields	Invalid or inconsistent data entry

7.4 Governance and Administration

The audit trail of Zoho Creator records who added or changed a record and when. A naming convention is followed: form names and field link names use words joined by underscores (for example Crime_Report_Form and Assignment_Officer). Data can be exported from reports as a backup. The Administrator manages user licences, profiles and sharing rules, and is responsible for maintenance, version notes and removing access for users who have

left the department. Records of closed cases should be kept according to the retention rules of the police department.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The completed application provides eight forms, eight reports and one dashboard. The table below lists the main outputs and the figures in which they are shown.

Table 8.1: Application Outputs and Where They Are Shown

Output	Description	Shown In
Admin_Dashboard	Two donut charts, one bar chart, embedded Assign Crime Case form and Crime Reports Register	Figures 5.1, 5.5
Crime Report Form	Complaint entry with crime type, location address, description and evidence files	Figure 5.2
Case Assignment form	Assignment of a report to an officer using picklist references	Figure 5.3
Crime_Report_Form_ Report	List of complaints with their statuses	Figure 5.4
Profiles and sharing rules	Access configuration for Administrator, Police Officer and Citizen	Figure 5.6
Phone and tablet layouts	Two-column phone layout and sliding-pane tablet layout	Figure 5.7

8.2 Achievement of Objectives

Table 8.2: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Data model of eight connected forms	Sections 4.2, 4.3, 5.2	Eight forms with 65 fields; Figure 4.2
Picklist references between forms	Section 4.2, Table 4.1	Ten picklist references to reports, officers and citizens
Location capture and evidence upload	Section 5.2	Address coordinate capture; two file-upload fields (10 files each)
Eight reports	Section 5.3, Table 5.1	One report for every form
Admin_Dashboard	Section 5.4, Table 5.2	Three charts, one embedded form, one embedded report
Profiles, roles and sharing rules	Sections 4.6, 5.7	Seven profiles, three roles, five sharing rules
Navigation and device support	Sections 4.5, 5.10	Space menu; web, phone and tablet layouts

8.3 Comparison with the Existing System

Table 8.3: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording a complaint	Paper register, then copied to files	Online form with automatic Report_ID	Faster, no copying
Linking records	Not linked; manual search	Picklist references between forms	Easy tracing of a case
Evidence tracking	Handwritten notes and folders	Evidence form with type, status, files and storage reference	Better traceability
Case statistics	Counted by hand	Dashboard charts by type and status	Always up to date
Closure record	Written in register	Case Resolution with approver and closure status	Clear accountability
Citizen feedback	Rarely collected	Rating, comments and suggestions	Service quality measured
Security	Open registers and files	Profiles, roles and sharing rules	Data protected

The comparison is based on the working of the manual process and the features of the application; it should be supported with measured timings if available.

8.4 Limitations of the Low-Code Approach

Some limits were noticed while building the application. The look of the forms can be changed only within the options given by the platform. The application depends on Zoho, so moving it to another platform would need rebuilding. A crime report system holds sensitive information, so the department must check that hosting data on a public cloud follows its rules. In the present version the complaint is not linked directly to a registered citizen, and there is no automatic notification or automatic status change because no scripts or workflows were used. These limits can be reduced by adding Deluge automation and connections in future.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

This project used the **no-code to low-code** end of the spectrum: forms, reports, dashboard, access control and layouts were all configured visually. The result shows that a useful business application with eight connected forms can be created by one student in a short time without writing code. For features such as automatic assignment, SMS alerts or integration with an official database, the project can move to the **low-code** level with Deluge scripts and finally to the **pro-code** level with custom functions, widgets or the Creator API. The platform therefore allows an application to grow as its requirements grow.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Crime Report Management System was developed on Zoho Creator to manage crime complaints from submission to resolution. Eight connected forms store citizens, crime reports, police officers, case assignments, investigations, evidence, case resolutions and citizen feedback. Picklist references connect the records, eight reports give access to the data, and the Admin_Dashboard shows crime types and case status with charts and gives quick access to case assignment and the crime report register. Profiles, roles and five sharing rules protect the information, the Space menu organises the modules, and web, phone and tablet layouts make the application usable on different devices.

The project replaces paper registers and scattered files with one online system, reduces repeated data entry, makes cases traceable through the Report_ID, and gives administrators a live view of the work. It also shows how quickly a complete application can be produced with a low-code platform.

9.2 Future Enhancements

- Link Crime_Report_Form directly to Citizen_Registration so that a complaint is connected to a registered citizen.
- Add Deluge form workflows to update Report_Status automatically when an assignment, investigation or resolution is saved.
- Send e-mail or SMS notifications to the citizen and to the assigned officer.
- Add an approval process for case closure, so that the Closure value is approved by a senior officer.
- Add map reports and heat maps using the captured latitude and longitude to show crime hot-spots.
- Add schedule workflows to remind officers of pending cases and overdue investigations.
- Use Zia or AI services for crime-pattern analysis, duplicate-complaint detection and summarising investigation notes.
- Integrate with an official records system through the Zoho Creator REST API and publish a citizen portal.

9.3 Learning Outcomes and Career Relevance

Through this project I learned how to analyse a real-world process and convert it into a data model; how to create forms, fields, picklist references, reports and dashboards in Zoho Creator; how to control access with profiles, roles and sharing rules; how to design for web, phone and tablet; and how to plan testing, deployment and governance. These skills are

useful for roles such as low-code developer, business application developer, Zoho Creator consultant and business analyst, and they provide a base for learning Deluge scripting and integration.

REFERENCES

[1] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/

[2] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/

[3] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/

[4] A. C. Bock and U. Frank, “Low-Code Platform,” Business & Information Systems Engineering, vol. 63, no. 6, pp. 733–740, 2021.

[5] R. Sanchis, Ó. García-Perales, F. Fraile and R. Poler, “Low-Code as Enabler of Digital Transformation in Manufacturing Industry,” Applied Sciences, vol. 10, no. 1, p. 12, 2020.

[6] A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA), 2020, pp. 171–178.

APPENDIX A
ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = done, P = partly done, N = not used.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Table
I	Application created and sections / menu organised	Y	Space menu with 12 sections	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	All eight forms (10 field types)	5.2	Fig. 5.2
II	Picklist reference (lookup) between forms	Y	10 references: report, officer, citizen	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	N	Not used; separate forms are linked by references	4.2	—
II	Field restrictions / hidden fields	P	Picklist-restricted values, hidden latitude and longitude, 10-file limit	5.2	Fig. 5.2
II	List report	Y	Eight reports; embedded report with bulk actions disabled	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline / Map report	N	Not used	5.3	—
II	Pivot table / chart report	P	Donut and bar charts on the dashboard (no separate chart report)	5.4	Fig. 5.5
II	Page / dashboard	Y	Admin_Dashboard	5.4	Fig. 5.5
II	Embedded form and report in a dashboard	Y	Assign Crime Case and Crime Reports Register	5.4	Fig. 5.5
II	Deluge – form events, functions, fetch, loops	N	Not used (see 5.5)	5.5	—
II	Form workflow / schedule / batch workflow	N	Not used (see 5.5)	5.5	—
II	Email / in-app notifications	N	Not used (future enhancement)	5.5, 9.2	—
II	Approval process / Blueprint	N	Approvals menu section present; no custom process configured	5.6	—
II	Testing and debugging evidence	P	Test cases and approach prepared; execution results to be recorded	6	Fig. 6.1
III	Integration with a Zoho service / REST API / webhook	N	Not used (see 5.6)	5.6	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Table
III	Users, roles and permission profiles	Y	Seven profiles; roles CEO, Police Officer, Citizen	4.6, 5.7	Fig. 5.6
III	Record-level access (sharing rules)	Y	Five Read-only sharing rules	4.6, 5.7	Table 4.11
III	Zia / AI feature	N	Planned (see 5.8)	5.8	—
III	Analytics (charts)	Y	Three dashboard charts	5.4, 5.8	Fig. 5.5
IV	Performance optimisation	P	Design choices applied; measurement pending	7.2	Table 7.2
IV	Deployment across environments	P	Strategy defined; production publish to be confirmed	7.1	Fig. 7.1
IV	Security controls	Y	Profiles, roles, sharing rules, hidden fields	7.3	Table 7.3
IV	Governance (audit trail, naming convention)	P	Built-in audit trail; naming convention followed	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used (see 5.9)	5.9	—
V	Low-code limitations and spectrum analysis	Y	Discussion in Chapter 8	8.4, 8.5	—
—	Mobile / tablet access	Y	Phone and tablet layouts for all eight forms	5.10	Fig. 5.7

Total features implemented: 12 fully and 6 partly out of 27.

APPENDIX B
COURSE OUTCOME MAPPING

Table A.2: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.5), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.6–5.8		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.8, 5.9, 8.4, 8.5, 9.3		

APPENDIX C
APPLICATION CONFIGURATION SUMMARY

The application has no custom Deluge source code, so this appendix summarises the configuration of the application instead.

C.1 Forms, Fields and References

Table C.1: Summary of Forms

S.No.	Form (Link Name)	Fields	Picklist References	File Upload	Matching Report
1	Citizen_Registration	7	0	No	Citizen_Registration_ Report
2	Crime_Report_Form	11	0	Yes (10 files)	Crime_Report_Form_ Report
3	Police_Officer	7	0	No	Police_Officer_Report
4	Case_Assignment	7	2	No	Case_Assignment_Report
5	Investigation_Details	8	2	No	Investigation_Details_ Report
6	Evidence_Management	10	2	Yes (10 files)	Evidence_Management_ Report
7	Case_Resolution	8	2	No	Case_Resolution_Report
8	Citizen_Feedback	7	2	No	Citizen_Feedback_Report
	Total	**65**	**10**	**2 fields**	**8 reports**

C.2 Picklist Option Values

Table C.2: Picklist Fields and Their Options

Form	Field	Options
Citizen_Registration	Gender	Male, Female, Transgender
Crime_Report_Form	Crime_Type	Theft, Robbery, Cybercrime, Assault, Missing Person, Fraud, Property Damage, Other
Crime_Report_Form	Report_Status	Submitted, Under Review, Under Investigation, Resolved, Closed
Police_Officer	Rank	Inspector, Sub-Inspector, Head Constable, Constable, Other
Case_Assignment	Priority	Low, Medium, High
Case_Assignment	Assignment_Status	Assigned, In progress, Completed
Investigation_Details	Investigation_Status	Not Started, In Progress, Evidence Collected, Completed
Evidence_Management	Evidence_Type	Photograph, Video, Document, Fingerprint, Physical Evidence, Digital Evidence, Other
Evidence_Management	Evidence_Status	Collected, Under review, Verified

Form	Field	Options
Case_Resolution	Final_Outcome	Suspect Arrested, Property Recovered, Complaint Resolved, Insufficient Evidence, Case Referred, Other
Case_Resolution	Closure	Pending Approval, Approved, Rejected, Closed
Citizen_Feedback	Rating	1 – Very Poor, 2 – Poor, 3 – Average, 4 – Good, 5 – Excellent

C.3 Dashboard, Profiles and Sharing Settings

- **Dashboard:** Admin_Dashboard with Crime Reports by Type (donut, Crime_Type, Count of ID), Cases by Status (donut, Report_Status, Count of ID), Cases by Status (bar, Report_Status, Count of ID), embedded form Assign Crime Case (Case_Assignment) and embedded report Crime Reports Register (Crime_Report_Form_Report).
- **Profiles:** Administrator, Police_Officer_Permissions, Citizen_Permissions, Read, Write, Developer, Customer.
- **Roles used in sharing rules:** CEO, Police Officer, Citizen.
- **Sharing rules:** Share_Investigation_Details_With_Police, Share_Evidence_With_Police, Share_Case_Assignments_With_Police, Share_Crime_Reports_With_Police, Share_Case_Resolutions_With_Admin (all Read).
- **Menu:** Space (Crime Report Form, App Preferences, Approvals, Reports, Citizen Registration, Police Officer, Case Assignment, Investigation Details, Evidence Management, Case Resolution, Citizen Feedback, Admin Dashboard).
- **Layouts:** Web (Lato font, left-aligned labels, icon text CRM); Phone (automatic labels, two columns, Chrysler Purple); Tablet (automatic labels, sliding pane, Dark Grey).

APPENDIX D
USER MANUAL

D.1 Accessing the Application

Open the application link [paste the live URL here] in Chrome or Edge, or open the Zoho Creator mobile app. Sign in with the Zoho account given by the Administrator. The Space menu on the left shows the sections that are allowed for your profile.

D.2 Instructions for the Administrator

- Open Police Officer and add each officer with name, badge number, rank, police station and contact details.
- Open the Admin Dashboard to see the charts of crime reports by type and cases by status.
- Use the Assign Crime Case form on the dashboard: select the Report_ID and the officer, enter the date, priority and instructions, and click Assign Case.
- Use the Crime Reports Register on the dashboard to view complaints and their statuses.
- Use Settings → Users and Control to add users and assign the Administrator, Police_Officer_Permissions or Citizen_Permissions profiles.
- Open Case Resolution Report to view the resolutions recorded by police officers.

D.3 Instructions for the Citizen

- Open Citizen Registration and enter your name, date of birth, gender, mobile number, e-mail and address, then click Submit.
- Open Crime Report Form, enter the complainant details, select the crime type, enter the incident date, time and location, describe the crime and upload evidence files (up to 10). Click Submit and note the Report_ID.
- After the case is handled, open Citizen Feedback, select the Report_ID and your name, choose a rating, and enter comments and suggestions.

D.4 Instructions for the Police Officer

- Open Crime Report Form Report and Case Assignment Report to see the reports and cases given to you.
- Open Investigation Details, select the Report_ID and your name, and enter the date, findings, action taken, status and remarks.
- Open Evidence Management, select the Report_ID, choose the evidence type, enter the description, upload files, select the collecting officer, enter the storage reference and set the evidence status.

- Open Case Resolution, select the Report_ID, enter the date, final outcome, summary, approving officer and closure status, and click Submit.